25901924 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA202 - DATA ANALYSIS IN INFERENTIAL STATISTICS USING R/PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. State the formula for the conditional probability P(Y=y|X=x) in terms of the joint and marginal probabilities. [K] / [CO1]

2. Given the joint probability mass function as $f(x,y) = \frac{xy^2}{30}; x = 1,2,3; y = 1,2.$ Find the marginal probability mass function of Y. [U] / [CO1]

3. State the pdf of the sampling distribution of mean of a sample of size *n* from a normal distribution. [K] / [CO2]

4. If t ~ t_{14} , Find P(t > 2.262). [A] / [CO2]

5. State any two applications of the F-distribution in statistical analysis. [U] / [CO2]

6. What does 99% confidence level mean? [U] / [CO3]

7. What are the properties of a good point estimator? [K] / [CO3]

8. Define Type II Error. [U] / [CO4]

9. What are assumptions of t-test? [U] / [CO4]

10. In a one way ANOVA, what does the term 'one-way' refer to? [U] / [CO4]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. If f(x, y) = cxy; x = 0, 1 and y = 1, 2 is a joint pdf, find (i) c (ii) P(X = 1, Y = 2), (iii) P(Y=2). [U] / [CO1]

12. Discuss different inter-relations among sampling distributions. [An] / [CO2]

13. Explain sampling distributions with suitable examples. [K] / [CO2]

14. Define the concept MLE and state any three of its important properties. [K] / [CO3]

15. A sample of 100 students is taken from a population with heights having s.d. 10 cm. the mean height of the sample of students is found to be 168.8cm. can we accept the assumption that the mean height of the students is 170 cm? [A] / [CO4]

16. A survey finds 21% of 222 men and 18% of 208 women purchased books online. Check whether men are more likely to purchase books online at α = 0.05? [A] / [CO4]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. What does the value of the joint distribution function at a point (x,y) represent? Using the following situation explain the concept. A fair coin is tossed 3 times. Let X be the number of heads in the first two tosses and Y be the number of heads in all three tosses. [U] / [CO1]

18. a. Explain the concept of the Chi-Square test for homogeneity of attributes. [A] / [CO4]

    b. A study was conducted to examine whether people in three cities are

homogeneous with respect to their opinion (Support, Neutral, Oppose) on a new transport policy. The results are as follows:

City	Support	Neutral	Oppose	Total
City A	40	30	30	100
City B	50	20	30	100
City C	30	40	30	100
Total	120	90	90	300

Test at 5% level of significance whether the cities are homogeneous in their opinion.

[12x1 = 12]